assignment impulse-momentum theorem

This is a preview of the published version of the quiz

Started: Apr 29 at 7:59pm

Quiz Instructions

No tutorial but hints (hide them first)

Question 1 9.1 pts

The average braking force of a 1000-kg car moving at 10 m/s braking to a stop in 5 s is

- 3000 N.
- 1000 N.
- 5000 N.
- 2000 N.
- 4000 N.

Question 2 9.1 pts

A 45.0-kg person steps on a scale in an elevator. The scale reads 460 N. What is the magnitude of the acceleration of the elevator?

hint: Find the peron true weight (mass x 9.8). If the apprent weight (N=460N) is larger than the true weight, the elevator is accelerating say @ up. Otherwise, its accelerating @ down. See slides. Draw the forces acting on the person. if its going up. N - weight = ma (up - down = ma) . solve for a. Otherwise, if its accelerating down: weight - N = ma. First decide it its @ up or @ down.

- 9.81 m/s^2
- 0.206 m/s^2
- 4.91 m/s^2
- 46.9 m/s^2

○ $0.422\ m/s^2$

Question 3 9.1 pts

A karate expert executes a swift blow and breaks a cement block with her bare hand. The magnitude of the force on her hand is

hint: remember Newton's third law

○ the same as the force applied to the block.

○ more than the force applied to the block.

○ need more information

○ zero.

○ less than the force applied to the cement block.

Question 4 9.1 pts

When a boxer is moving away from a punch, the force experienced is reduced because

○ all of the above

○ the time of contact is increased.

○ momentum transfer is reduced.

○ the force is less effective.

Question 5 9.1 pts

A 0.140-kg baseball is dropped and reaches a speed of 1.20 m/s just before it hits the ground and bounces. It rebounds with an upward velocity of 1.00 m/s. What is the change of the ball's momentum during the bounce?

hint: change in momentum = mass (final velocity - initial velocity). velocity is a vector. down means negative. Change in momentum is a vector too.

○ 0.308 kg · m/s upwards

○ 0.0280 kg · m/s downwards

○ 0.308 kg · m/s downwards

○
$0.000\ \mathrm{kg} \cdot \mathrm{m/s}$

○
$0.0280\ \mathrm{kg} \cdot \mathrm{m/s}$ upwards

Question 6 9.1 pts

A karate chop is more effective if one's hand

○
extends the time upon impact.

○
follows through upon impact.

○
bounces upon impact.

Question 7 9.1 pts

When a boxer moves into an oncoming punch, the force experienced is

○
decreased.

○
increased.

○
no different, but the timing is different.

○
all of the above

Question 8 9.1 pts

It is correct to say that impulse is equal to

hint: see slides/ Definitions

○
momentum.

○
velocity multiplied by time.

○
a corresponding change in momentum.

○
force multiplied by the distance it acts.

Question 9 9.1 pts

The impulse-momentum relationship is a direct result of

hint: see slides.

○
Newton's law of gravity.

Newton's 3rd law.

Newton's 1st law.

Newton's 2nd law.

Question 10 9.1 pts

To catch a fast-moving ball, you extend your hand forward before contact with the ball and let it ride backward in the direction of the ball's motion. Doing this reduces the force of contact on your hand principally because the

none of the above

relative velocity is less.

time of contact is decreased.

force of contact is reduced.

time of contact is increased.

Question 11 9.1 pts

Compared with falling on a stone floor, a wine glass may not break when it falls on a carpeted floor because the

stopping time is longer on the carpet.

carpeted floor provides a smaller impulse.

carpet provides a smaller impulse and a longer time.

stopping time is shorter on the carpet.

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